Interpretable Deep Learning For Automated Retinal Damage Detection in Optical Coherence Tomography

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***Abstract*— This work portrays developing and evaluating deep learning models for automatically identifying retinal injuries using Optical Coherence Tomography (OCT) images. In particular, this study optimizes three pre-trained Convolutional Neural Networks (CNNs), VGG16, EfficientNetB3, and ResNet50, using transfer learning to classify retinal images into four categories, Drusen, Normal, Diabetic Macular Edema (DME), and Choroidal Neovascularization (CNV). In addition to high-performance classification, the study integrates Explainable Artificial Intelligence (XAI) methods, utilizing Gradient-weighted Class Activation Mapping (Grad-CAM) to highlight the retinal regions influencing the model's predictions visually. This offers critical interpretability for clinical settings. The Key performance metrics indicators include accuracy, precision, and F1-score and help to evaluate the models. With the available selected models, EfficientNetB3 achieved the highest performance with an accuracy of 93%, outperforming VGG16 and ResNet50. The application of Grad-CAM highlights the increase in transparency and prediction trust of this model's decision-making process. This study demonstrated the profound learning model efficacy, particularly EfficientNetB3, for accurate retinal injury detection and provides valuable insights into model interpretability for clinical use.**

Keywords*— *Optical Coherence Tomography, deep learning, EfficientNetB3, Grad-CAM, diagnostic interpretability, Explainable AI

I. INTRODUCTION

One of the most important parts of the human eye is the retina, which is in charge of gathering and interpreting the visual data required for eyesight. It has a thin layer of tissue at the rear of the eye made up of rods and cones, which are specialised cells known as photoreceptors. We are able to perceive our surroundings because these cells are able to detect light and transform it into neural signals that are transmitted from the optic nerve to the brain. The retina's health is important for preserving clear vision, and any damage or disease affecting this fragile structure can lead to serious vision impairment or blindness.

If left untreated, a number of common retinal disorders, namely Drusen, Diabetic Macular Oedema (DME), and Choroidal Neovascularization (CNV), can have a significant negative influence on vision. For instance, CNV is typified by the aberrant development of blood vessels behind the retina, which may result in bleeding and scarring and swiftly impair vision. DME is a disorder that happens when fluid builds up in the macula as a result of blood vessel leaking, which is frequently linked to diabetes. Yellow deposits under the retina called drusen are often associated with age-related macular degeneration (AMD), and their presence can indicate the disease's early stages. In order to diagnose and treat these conditions, we use Optical Coherence Tomography (OCT) images of each disease that exhibit distinctive characteristics.

A non-invasive imaging technique called optical coherence tomography (OCT) produces cross-sectional images of the retina with great resolution. With the use of this cutting-edge imaging method, doctors can see the retina's layers in great detail, which helps in the early identification and monitoring of retinal illness. OCT has gained widespread use in ophthalmology as a diagnostic tool for diseases including CNV, DME, and AMD because of its capacity to offer precise structural information about the retina. Due to its ability to pick up on even the smallest variations in retinal thickness and structure, it is invaluable for spotting pathological changes that conventional fundus photography could miss.

OCT imaging provides valuable insights, but manual interpretation is challenging due to the volume and complexity of retinal structures. Automated OCT image analysis is crucial for early detection and treatment, especially in large-scale screening programs. The global prevalence of retinal diseases, especially in aging populations, necessitates efficient detection methods. Automated detection systems can enhance clinical workflows and alleviate the global burden of preventable blindness.

II. LITERATURE SURVEY

In this section, we conduct a comprehensive review of the existing literature related to Retinal Damage Detection in Optical Coherence Tomography. By synthesizing these findings, we aim to provide contextual foundation for the proposed methodology and underscore the necessity for continued innovation in this domain.

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Deep learning (DL) has made substantial advances in illness identification and diagnosis possible for computer-aided diagnostic systems (CADS) [1] by improving accuracy, resilience, and efficiency. The goal of this field's research has been to create automated artificial intelligence (AI) technologies that can diagnose internal bleeding, fractures, tumors, preoperative planning, and intraoperative guiding with the least amount of human intervention. However, there are a number of obstacles that prevent the use of DL this diagnosis. One major drawback is the dearth of balanced, well-annotated medical imaging datasets, which are necessary for building trustworthy models. Furthermore, noise in medical imaging data and adversarial attacks might compromise the diagnostic accuracy of deep neural networks (DNNs).

A study investigating the use of Optical Coherence Tomography (OCT) [2] for real-time ablation process monitoring in the context of laser osteotomy has focused on improving precision while minimizing tissue damage. OCT is especially useful for differentiating between different tissue types, including nerves and bone marrow, which gives the ablation laser vital feedback. The inherent noise in OCT images is a common difficulty that might reduce the accuracy of tissue classification. Deep learning-based denoising approaches have recently been offered as an alternative to traditional methods like frame-averaging, with equal quality.

In the field of Optical Coherence Tomography (OCT) [3] image denoising, traditional methods have faced significant limitations due to the prevalence of speckle noise, which hinders image quality. Deep learning has emerged as an effective tool for denoising OCT images, but its development has been constrained by the difficulty in acquiring large, well-registered paired OCT datasets. To address this, unpaired learning methods are used, most notably cycleGAN-based models. These models require dual generators and discriminators during training, leading to high computational demands that may be excessive for denoising tasks. In response, a novel approach called Triplet Cross-Fusion Learning (TCFL) has been proposed.

Identifying malignant spots quickly and accurately during surgery is essential for improving patient outcomes in the field of surgical oncology [4]. This research suggests a novel two-stage architecture that combines segmentation and image generation to address the problems of dynamic cell imaging (DCI) which includes color volatility, artefacts between patches in whole-slide photos, and a lack of precise annotated data pose difficulties for deep learning-based segmentation of DCI images. The Dual Semantic Diffusion Model (DSDM), intended to produce superior, semantically relevant DCI images, is presented. In order to improve feature extraction, further steps are taken to repurpose the pretrained DSDM to extract diffusion features.

Pancreaticobiliary carcinomas, a highly malignant gastrointestinal tumor, often arises from pancreatic intraepithelial neoplasias (PanINs) [5], which can progress to invasive duct adenocarcinoma (IDAC). However, a standard diagnostic protocol for detecting the transition from PanINs to IDAC remains elusive. In response to this gap, this study introduces a tri-modality imaging system integrating ultrasound (US), optical coherence tomography (OCT), and near-infrared fluorescence (NIRF) for real-time imaging of the pancreaticobiliary ducts. This system, capable of acquiring images at 30 frames per second, features a compact endoscopic probe with an outer diameter of 0.9 mm, combining a US transducer and a fiber ball lens for optimal imaging.

Drusen segmentation in retinal OCT images [6] is essential for early diagnosis of Age-related Macular Degeneration (AMD) [6] , but it faces challenges such as variability in drusen size and shape, blurred boundaries, and speckle noise. To address these issues, a novel multi-scale transformer global attention network (MsTGANet) is proposed, incorporating a multi-scale transformer non-local (MsTNL) module to capture long-range dependencies and a multi-semantic global channel and spatial attention module (MsGCS) to improve feature fusion. A semi-supervised version (Semi-MsTGANet) is also introduced to leverage unlabeled data via pseudo-labeling.

Optical Coherence Tomography (OCT) images [7] are often affected by speckle noise, which obscures critical tissue details and hinders clinical diagnosis. To address this, a novel tensor ring decomposition-guided dictionary learning (TRGDL) model is proposed for OCT image denoising. It combines low-rank and sparsity priors under a unified framework and enforces tensor ring decomposition and shared dictionary learning to enhance denoising performance. Experimental results demonstrate that TRGDL significantly outperforms existing methods, offering improved denoising for OCT images.

Retinal layer segmentation in OCT images [8] face challenges such as noise, blurring, and tissue artifacts, leading to false positives and boundary deviations. To address these issues, the TCCT-BP method is introduced, combining CNN and lightweight Transformer architectures for enhanced layer perception. Additionally, feature polarization and boundary regression loss functions are implemented to differentiate retinal layers and improve boundary accuracy.

Deep Learning (DL) has transformed healthcare applications like diagnostics but lack uncertainty estimation, which can lead to overconfident predictions [9]. To address this, Bayesian Deep Learning (BDL) was introduced, providing a probabilistic approach to quantify uncertainties in predictions. This study evaluates BDL techniques, their integration with architectures like CNNs and RNNs, and their applications in medical imaging, signal processing, and electronic health records.

Traditional MIL methods often assume definitive bag labels, neglecting the ambiguity inherent in this labeling process. Multi-instance learning (MIL) [10] typically associates labels with bags rather than individual instances, leading to challenges when bags are labeled with weak labels due to unknown labeler weights. This study introduces the Transfer Learning-based Multiple Instance Learning (TMIL) framework to address challenges by facilitating knowledge transfer between source and target tasks, both containing weak

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labels. The TMIL model iteratively updates bag labels to enhance MIL performance, demonstrating convergence and outperforming existing MIL methods by correcting initial weak labels to achieve more accurate representations.

III. Proposed Methodology

Figure 1 represents a novel workflow of the proposed OCT (Optical Coherence Tomography) image classification process. The process begins by dividing the OCT dataset into training and test sets. After this, image pre-processing is applied to prepare the images for model input. A ResNet50 model is loaded initially, followed by transfer learning using the VGG16 model to leverage pre-trained weights for enhanced performance. The model is then trained using the training set and tested on the test set. The final step is to classify the images into four categories: CNV, DME, DRUSEN, and NORMAL. The classified images are further visualized using Grad-CAM, which helps identify the important regions in the image that influenced the model's decision.

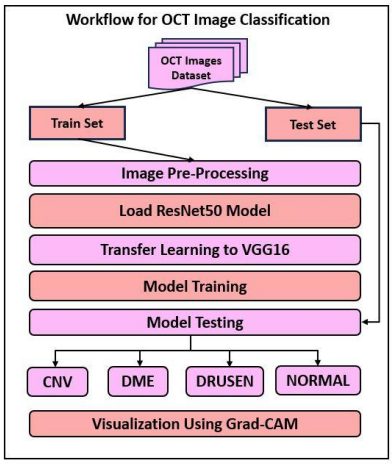


Fig. 1. Workflow for OCT Image Classification.

The architecture of the deep learning model for the OCT image classification is presented in Figure 2. Three colour channels (RGB) and 150x150 pixel images make up the classification input. To minimise the spatial dimensions and capture significant characteristics, the architecture consists of many convolutional layers with increasing filter sizes (64, 128, 256, and 512) interspersed with max-pooling layers. Following flattening and passing through a dense layer, the output of the last convolutional layer is classified into one of the following four groups: CNV, DME, DRUSEN, or NORMAL. In order to facilitate precise categorisation, this architecture aims to efficiently extract significant features from the OCT images.

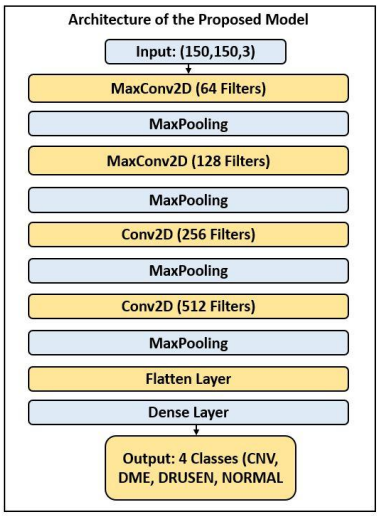


Fig. 2. Architecture of Proposed Model.

A. Dataset and Pre-processing

Images from four different classes—Normal, Diabetic Macular Oedema (DME), Choroidal Neovascularisation (CNV), and Drusen—make up the OCT dataset used in this investigation. A total of 4,000 images, publicly available on Kaggle were used, with 1,000 from each class. In order to guarantee consistency and model compatibility, pre-processing was performed. The first step was applying picture normalization to all photos in order to standardize the pixel values and ensure that the model was given inputs with consistent intensities. The photos were then resized to 224 x 224 pixels in resolution to comply with the models' criteria for input size.

B. Model Selection: VGG16, EfficientNetB3, and ResNet50

Three state-of-the-art deep learning models—VGG16, EfficientNetB3, and ResNet50—were selected for this study. Transfer learning was used to modify VGG16, which was pre-trained on ImageNet. The weights from all the layers of VGG16, tailored to capture retinal-specific features from OCT images, were transferred to the EfficientNet model through transfer learning. The early layers of the combined model, responsible for extracting low-level details like edges and textures, were kept frozen and their parameters remained unchanged during training. EfficientNetB3 was chosen to achieve optimal performance due to its effective architecture with fewer parameters, striking a compromise between computational resource utilization and performance. The model's simultaneous adjustments to depth, width, and resolution in scaling techniques made it possible to extract optimal features from the retinal image dataset. ResNet50 was selected due to its depth and capacity to solve the vanishing gradient issue. This enabled ResNet50 to effectively capture complex hierarchical features present in the OCT images, which were particularly useful for distinguishing between subtle retinal abnormalities.

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C. Training Setup

In order to ensure effective learning and optimise model performance, the training procedure was meticulously curated. With a learning rate of 0.0001, the Adam optimiser was used, which provided an adaptive method of weight alterations that accelerated convergence. Since it is ideal for multi-class classification tasks and ensures appropriate penalisation of inaccurate predictions across the four classes, the categorical cross-entropy loss function was employed. To avoid overfitting and ensure that the model did not train past the point of optimal generalisation, early stopping based on validation loss was applied for each of the 50 epochs of training that each model received. To keep steady training dynamics and memory efficiency in check, a batch size of 32 was employed. An NVIDIA Tesla V100 6GB GPU was used to train the models and it required up to 4GB of RAM during execution, providing the required computational power to handle the complex models and large dataset, significantly reducing the training time and improving performance. Training time averaged approximately 300-350 seconds per epoch, culminating in around 4100 seconds for complete model training across all epochs. For individual predictions with XAI-generated heatmaps, the model demonstrated efficient performance, requiring close to 3 seconds per classification. This indicates the capability of the model to perform well, particularly in low-resource environments.

D. Explainable AI with Grad-CAM

Grad-CAM (Gradient-weighted Class Activation Mapping) was used to further interpret the decisions made by the deep learning models. Grad-CAM produces heatmaps that show the precise areas in the OCT pictures that were most important to the model's predictions. This stage was crucial to confirming that the model was concentrating on the pertinent diseased regions of the retina, such as anomalies connected to diseases like Drusen or CNV (Choroidal Neovascularization). It was possible to verify that the model was examining significant features rather than deceptive picture artefacts by observing the crucial areas. This interpretability gave the model's decision-making process even more credibility, particularly in a delicate area like medical imaging.

These Grad-CAM-generated heatmaps are particularly useful for ophthalmologists and researchers in verifying the clinical accuracy of the model's outputs. For example, in cases where the model classified an image as CNV, the heatmap frequently emphasized regions showing abnormal growth of blood vessels, which is a hallmark of the condition. Similarly, for Drusen classification, the attention in the heatmap often centered on the yellow deposits beneath the retina, a defining feature of the disease.

IV. Results and Discussions

Figure 3 depicts normal retinal images, where the retinal layers appear smooth, without any significant abnormalities. These normal images act as a baseline for comparison with pathological conditions. OCT is a non-invasive imaging technique that provides cross-sectional views of the retina. Figure 4 represents the Drusen class, where small yellowish deposits called drusen can be seen beneath the retina, commonly associated with early stages of age-related macular degeneration (AMD). Figure 5 shows images from the CNV (Choroidal Neovascularization) class, where new abnormal blood vessels form under the retina, often leading to fluid leakage and scarring. This condition is associated with advanced stages of AMD and causes significant distortion in retinal structure. Figure 6 represents the DME (Diabetic Macular Edema) class, characterized by the accumulation of fluid in the macula due to leaking blood vessels, a common complication of diabetes that leads to swelling and vision impairment.

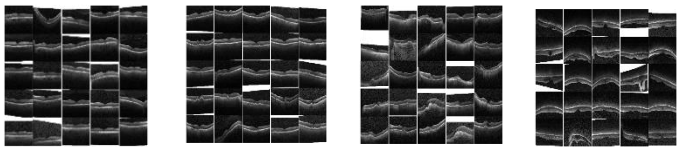


Fig 3,4,5 and 6. Samples from the dataset for all 4 classes

Figure 7 (a) and 7(b) illustrate the performance of the model, showing both accuracy and loss for the training and validation datasets over the course of several epochs. The left plot tracks model accuracy over time, with the training accuracy (blue line) increasing rapidly and reaching near-perfect levels around epoch 5. In contrast, the validation accuracy (orange line) also improves initially but levels off at around 90% after the fifth epoch, showing slight fluctuations. While the model continues to improve on the training data, its performance on unseen validation data stabilizes, indicating a potential point of minimal convergence.

The right plot depicts model loss, which measures the error in the model's predictions. Both the training loss (blue line) and validation loss (orange line) decrease over time, but the training loss consistently drops at a faster rate, reaching near-zero by the end of the training. The validation loss decreases more gradually and flattens after the initial epochs, suggesting that the model fits the training data well. This pattern often reflects a balance between optimizing the model and avoiding overfitting to the training data.

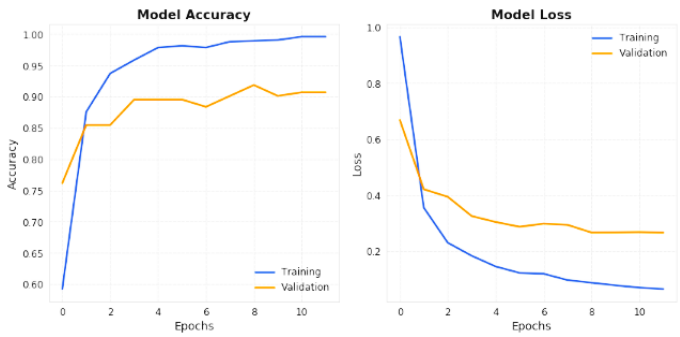


Fig 7(a) and (b). Performace Graphs during Training and Validation

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Figure 8 depicts a confusion matrix representing the classification performance of the model across four categories. The matrix displays a proportion of samples for each true-predicted class pair, with values ranging from 0 to 1. The diagonal elements show the model's correct classifications for each class: 90%, 98%, 86%, and 95% for the four respective categories. Off-diagonal values represent misclassifications, where a certain percentage of samples were incorrectly predicted as belonging to other classes. For instance, 7% of the samples from the Normal category were misclassified as the DME, and 2% as DRUSEN. This confusion matrix provides insights into the model's high accuracy and highlights the specific areas where misclassifications have occured, particularly between the first and third classes, indicating potential areas for improvement.

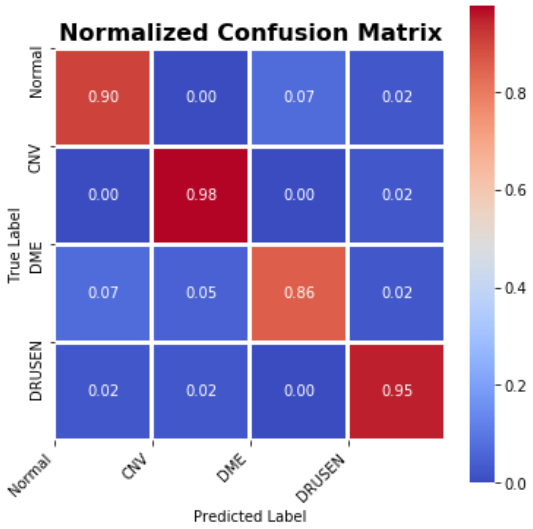


Fig 8. Normalized Confusion Matrix

Figure 9 depicts a radar chart that compares the accuracy of five state-of-the-art deep learning models used in our study: VGG16, Xception, EfficientNetB3, ResNet50, and InceptionV3. Each model is represented by a colored polygon, with accuracy values plotted along a radial axis ranging from 0.80 to 0.95, to give a visual representation of how these models perform relative to each other in terms of accuracy. The model represented by the red polygon, VGG16, exhibits a wider shape, indicating that its accuracy tends to be higher compared to the other models. In contrast, the models Xception, EfficientNetB3, and ResNet50, depicted by blue, green, and pink polygons respectively, touch just lower than 0.90 with overlap in their shapes. These models achieve similar levels of accuracy, clustering around the 0.85 to 0.90 range. InceptionV3, shown by the yellow polygon, displays the smallest and most compact shape, indicating that it achieves lowest accuracy among the models, with its values nearing the 0.80 mark.

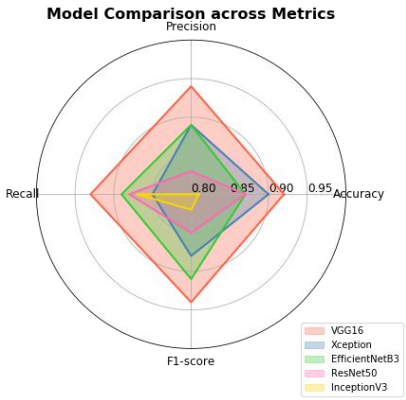


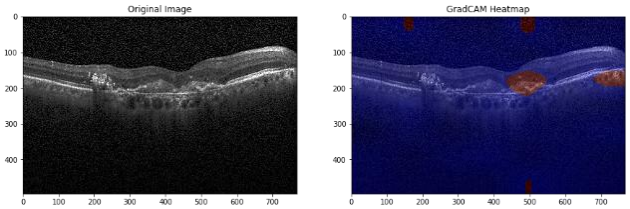
Fig 9. Model Accuracy Comparison Radar Chart for Various Architectures

To evaluate the effectiveness of the models used for OCT image classification, several performance metrics were considered, including accuracy, precision, recall, and F1-score. These metrics help in assessing not only how well the models perform overall but also their ability to correctly classify the different categories of retinal conditions (CNV, DME, Drusen, and Normal). A detailed table of these performance metrics for different models is shown below:

Table 1: Performance Metrics of the models

Model	Accuracy	Precision	Recall	F1-Score
ResNet50	87%	83%	88%	85%
VGG16	91%	89%	85%	88%
EfficientNetB3	93%	91%	90%	92%

The ResNet50 model exhibited strong performance with an accuracy of 87%, and demonstrating a balance between precision and recall, but still left room for improvement in handling false positives and false negatives. The VGG16 model showed better precision but a slight dip in recall, resulting in an overall improvement in F1-score compared to ResNet50. EfficientNetB3 emerged as the best-performing model, achieving 93% accuracy along with a precision of 91%, recall of 90%, and F1-score of 92%, making it the most reliable model in terms of classification accuracy and balance across all metrics.



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Fig 10. CNV - GradCAM Heatmap

Figure 10 shows an OCT scan of an eye affected by choroidal neovascularization (CNV) on the left, with its corresponding GradCAM heatmap on the right. The heatmap visualizes the attention of the model trained to detect CNV. The reddish regions in the heatmap indicate areas where the model has identified features indicative of CNV, which typically involve abnormal blood vessel growth under the retina. The gradual shift from blue to red signifies the varying levels of importance that the model assigns to different areas of the image. Red regions represent the highest attention, highlighting critical areas where the pathological changes are most pronounced.

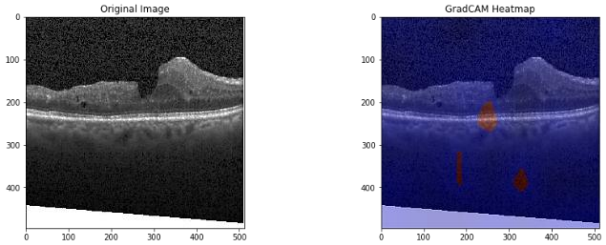


Fig 11 . DME - GradCAM Heatmap

Figure 11 represents an OCT scan of diabetic macular edema (DME) on the left, with its GradCAM heatmap on the right. DME is characterized by fluid accumulation in the macula, leading to retinal swelling. In the heatmap, the reddish regions correspond to the fluid-filled areas or abnormal retinal structures that are crucial for the model in identifying the presence of DME. The visualization shows that the model focuses on specific areas of pathology, confirming its ability to detect key features that distinguish DME from normal retinal conditions.

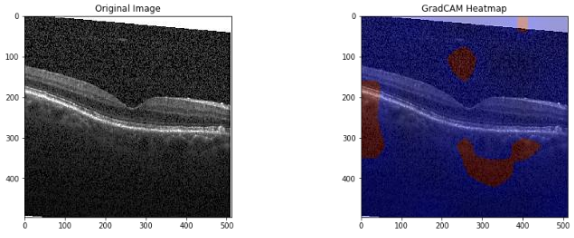


Fig 12. DRUSEN – GradCAM Heatmap

In Figure 12, the image on the left presents an OCT scan of an eye with drusen, a sign of age-related macular degeneration (AMD). The right side shows the Grad-CAM heatmap generated by the model. The model highlights drusen, which appears as reddish spots in the heatmap, indicating that this feature plays a significant role in its identification of the condition. The intensity and localization of the red zones show that the model is accurately focusing on the areas where drusen has accumulated between the retina and the underlying tissue, helping to differentiate it from other retinal diseases.

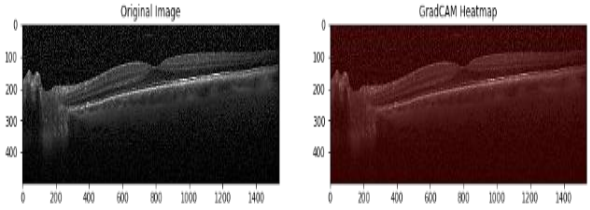


Fig 13. Normal – GradCAM Heatmap

Figure 13 depicts a normal OCT scan on the left, with the corresponding GradCAM heatmap on the right. Since the scan represents a healthy retina, the heatmap shows minimal attention from the model, with most of the image appearing in uniform color. This lack of focus in specific areas confirms that the model has correctly identified the absence of significant pathological changes. The uniform color across the heatmap indicates that no regions were flagged as important, signifying a successful distinction between normal and pathological retinal structures. Therefore, an Explainable AI technique was successfully applied for the visualization and interpretation of key insights in this study.

V. CONCLUSION AND FUTURE WORK

This research successfully applied deep learning models to detect retinal damage from Optical Coherence Tomography (OCT) images, leveraging pre-trained convolutional neural networks like VGG16, EfficientNetB3, and ResNet50 through transfer learning. The study focused on classifying OCT images into four categories: Normal, Drusen, Diabetic Macular Edema (DME), and Choroidal Neovascularization (CNV). Among the models tested, EfficientNetB3 emerged as the best performer, achieving a high accuracy of 93%, along with balanced precision, recall, and F1-scores. The application of Grad-CAM (Gradient-weighted Class Activation Mapping) allowed for greater interpretability of the model by highlighting critical retinal regions responsible for each classification, making the predictions more transparent and clinically relevant.

In addition to high classification accuracy, the study highlights the importance of XAI in making deep learning models interpretable for clinical settings. EfficientNetB3's ability to visualize the decision-making process via Grad-CAM makes it a promising tool for large-scale automated retinal screenings, aiding in the early diagnosis of retinal diseases such as age-related macular degeneration and diabetic macular edema. Future work could focus on integrating multimodal data, such as clinical and genetic information, to further improve diagnostic accuracy. Expanding the dataset to include a broader range of retinal conditions would also enhance the model's generalizability and clinical applicability.

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